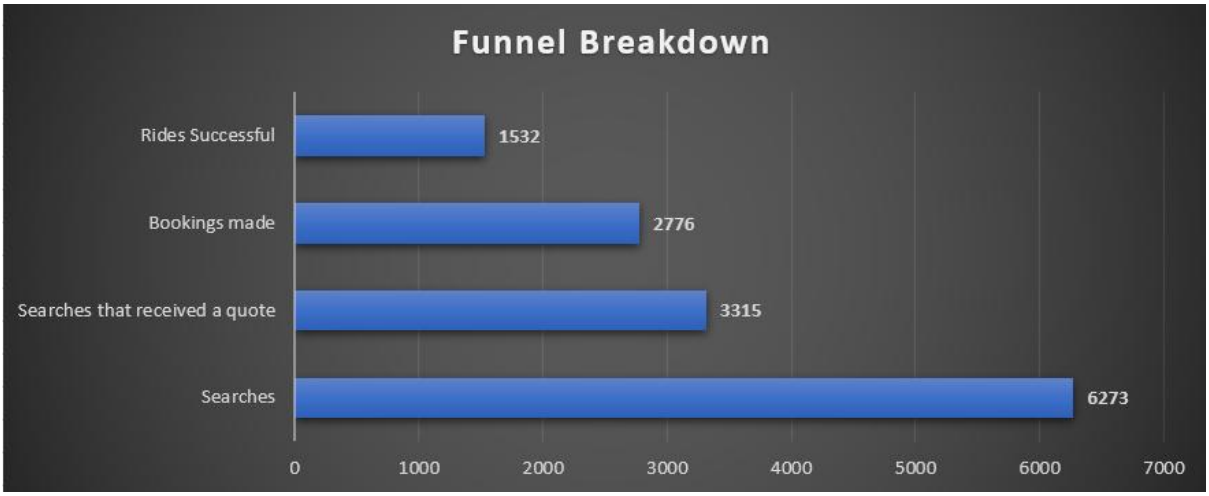
Namma Yatri Datathon Assignment

There are 4 csv files in the assignment folder. These are:

1. **Search_data.csv**: 6,273 rows. Each row is one ride search a rider made. The *estimated_distance* is in meters. The time stamp is in UTC, so if needed we can add +5:30 hours since Namma Yatri operates in India.
2. **Quote_data**.**csv**: 9621 rows, but multiple drivers can quote on one search. The link back to search is *search_request_id*. The *distance_to_pickup* is in meters.
3. **Booking_data**.**csv**: 2776 rows. One row per booking. It links to the quote via *quote_id.* The status column has 3 values: completed, cancelled, trip assigned.
4. **Booking_cancellation_data.csv**: 1263 rows, it links to the booking via *booking_id.* The source column tells us who cancelled: ByApplication, ByUser, ByDriver

Now, building the funnel from these files:



Using the data present in the excel files, we can clearly see that the biggest drop in the funnel happens at the very first stage: search to quote. **Out of 6,273 searches, only 3,315 received a quote** from a driver, **meaning 47.2% of riders never even got a response**. This points to a **supply-side gap where driver availability is not meeting rider demand**

Interestingly, **once the rider receives a quote, they almost proceed to book**. The **quote to booking** made **conversion is 83.7%,** suggesting that the **problem is not the fare acceptability**, or the **app experience at the booking stage**.

The final stage, from **booking made to ride successful** is the **second area of concern**, with only a **55.2% completion rate**, driven **primarily by the post bookings cancellations**.

Conversion rates across funnel stages:

Search to Quote	**52.8%**
Quote to Booking made	**83.7%**
Booking made to ride successful	**55.2%**
Search to Ride successful	**24.4%**

Time of the day:

Time Slot	Searches	Search -> Quote	Quote -> Booking	Booking -> Complete	End to End
Morning	1404	49.6%	84.6%	62.2%	26.1%
Day	1987	69.5%	83.3%	59.4%	33.9%
Evening	2528	42.3%	85.0%	46.5%	16.7%
Night	354	52.5%	76.9%	47.6%	19.2%

- The **evening period** is **where the most demand** is at **2528 searches** (40% of the total), it is also the period where the **search to quote** is **the worst** at **42.3%,** meaning **over half of evening riders never get a driver response**. This points to a clear **supply shortage during peak commute hours.**
- The **Day segment performs best end-to-end** at **33.9%,** but its **biggest drop** is **post-booking** (**59.4% completion**), suggesting **riders book and then cancel during working hours**.
- **Night trips** show the **weakest Quote to Booking rate at 76.9%,** possibly indicating **fare sensitivity** or **safety concerns late at night**.

Trip length:

Trip Length	Searches	Search -> Quote	Quote -> Booking	Booking -> Complete	End to End
Short	2093	41.6%	81.1%	47.7%	16.1%
Medium	3256	52.2%	86.0%	57.9%	26.0%
Long	924	80.5%	81.7%	57.4%	37.8%

- The **short trips**, have the **worst end to end conversion** at **16.1%,** with only **41.6% of searches receiving a quote.** This suggests drivers are deliberately avoiding short trips due to low fare economics, a driver travelling 2 km to pick up a rider for a 3 km trip earns very little relative to their fuel cost.
- Long trips above 15 km receive **nearly 4 quotes per search on average**, confirming **strong driver preference for higher-earning rides**. The **platform loses the most volume** at the **short-trip segment**, which likely represents a **large share of daily urban commuters**.

Overall Cancellation Rate:

- The **overall cancellation rate** is **44.7%,** meaning **nearly 1 in 2 bookings does not result in a completed ride**. This is the **single largest source of revenue leakage** on the platform. **Driver-initiated cancellations account** for **60.6% of all cancellations** (766 out of 1,263 recorded), making **driver behaviour the primary lever to address**.

Top Driver Cancellation Reasons:

- The **top 3 reasons** for **driver cancellations** are **NOT_REACHABLE (212 cases), PICKUP_LOC_TOO_FAR (141 cases),** and **VEHICLE_ISSUE (140 cases).**
- **NOT_REACHABLE peaks** during the **Day slot**, likely because **riders in office areas are in meetings or noisy environments**.
- **PICKUP_LOC_TOO_FAR** and **VEHICLE_ISSUE** both **peak during Evening**, **suggesting drivers are over-stretched during peak hours** and **accepting rides** they **cannot service**.
- Across **trip lengths**, all **three reasons are most frequent in medium trips**, which **make up the bulk of volume**.

Driver Rating vs Cancellation Behaviour:

- **Drivers who cancel** have a **median rating of 4.75** compared to **4.79 for non-cancelling drivers**, a difference of only 0.04 points. **This shows that star rating alone is a weak predictor of cancellation risk**.
- When broken down by rating bucket, the **4.5 to 5 rated drivers account for 76% of all cancellations** in absolute terms, **simply because most drivers fall in this range**.
- A **more reliable signal would be** tracking each **driver's accept-to-complete ratio over time** rather than relying on average ratings.

Analysis tool: Microsoft Excel. Supporting Excel file attached as proof of computation.